

8. PROGRAMS ON DAC INTERFACING

AIM: Develop necessary interfacing circuit to interface DAC with 8051 and program to generate different stair case pattern, ramp, triangular wave, sawtooth wave.

TOOLS REQUIRED: PC, Keil μ vision5

PROCEDURE:

1. Turn on the computer, create a folder on D drive saved with Register Number.
2. Open Keil uVision5 in desktop, or windows start menu → all programs → open Keil uVision5.

Creating Project:

3. Go to project → click on new uVision project → create a new folder saved with experiment number within the already existed register number folder in D drive mentioned in step 1, → enter the project name → click on save.
4. Select the device for target → In devices → Enter P89C51RD2XX in Search toolbar → click on ok → select **No** for dialog box message “Copy STARTUP.A51 to project folder and add files to project”.

(or)

Choose NXP → to select the device P89C51RD2xx → click on ok → select **No** for Copy STARTUP.A51 to the project folder.

Creating Coding File:

5. Go to file → click on new → go to save (choose the path to save the file, It is saved within the name of experiment number folder mentioned in step 3) → enter a filename with extension **.asm** → save the file.

Linking the Coding File to Project:

6. Right-click on Source group1 in project bar → Add existing files to source group1 → choose the experiment number folder path and select all files in the folder → select .asm code file → click on add → click on close.
7. Write the assembly language program in .asm code file and save it.

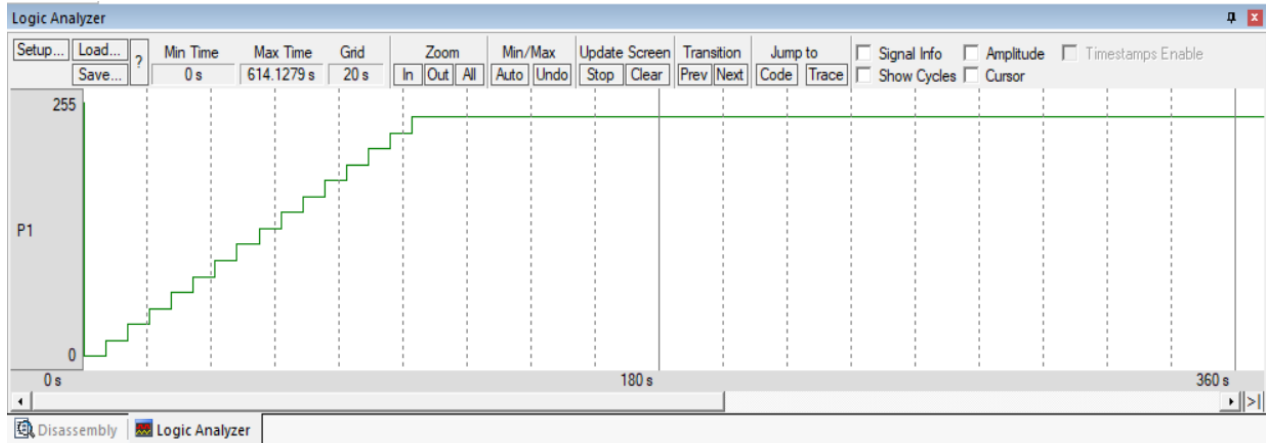
Executing the Code File:

8. Right-click on .asm code file → Click on Build target to check the errors (i.e 0-Errors, 0-Warning)
9. Go to debug → Click on Start/Stop Debug Session → click on ok for dialog box message “running code size limit 2K” → and Click on RUN in debug label
10. Observe the output in Register windows, Memory windows, Serial window.

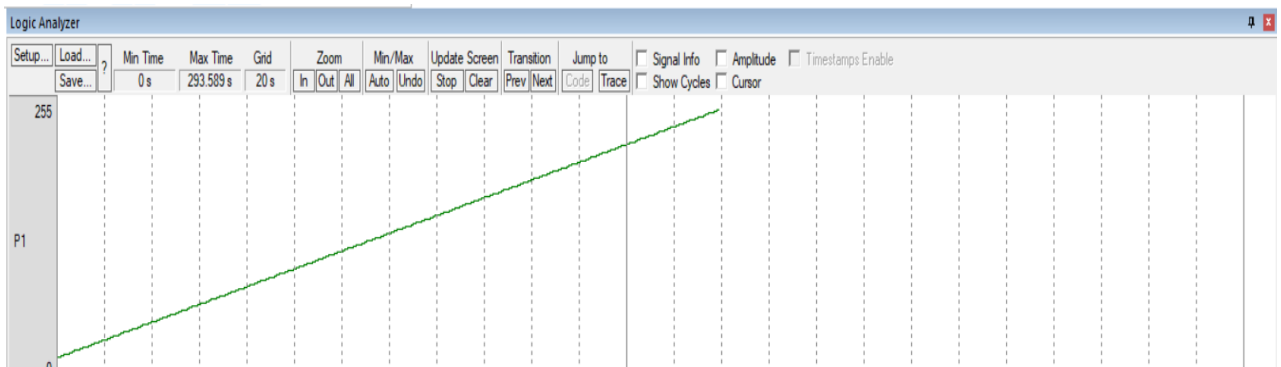
OUTPUT:

View → Analysis Windows → Logic Analyzer

1.STAIRCASE OUPUT:



2.RAMP OUTPUT:



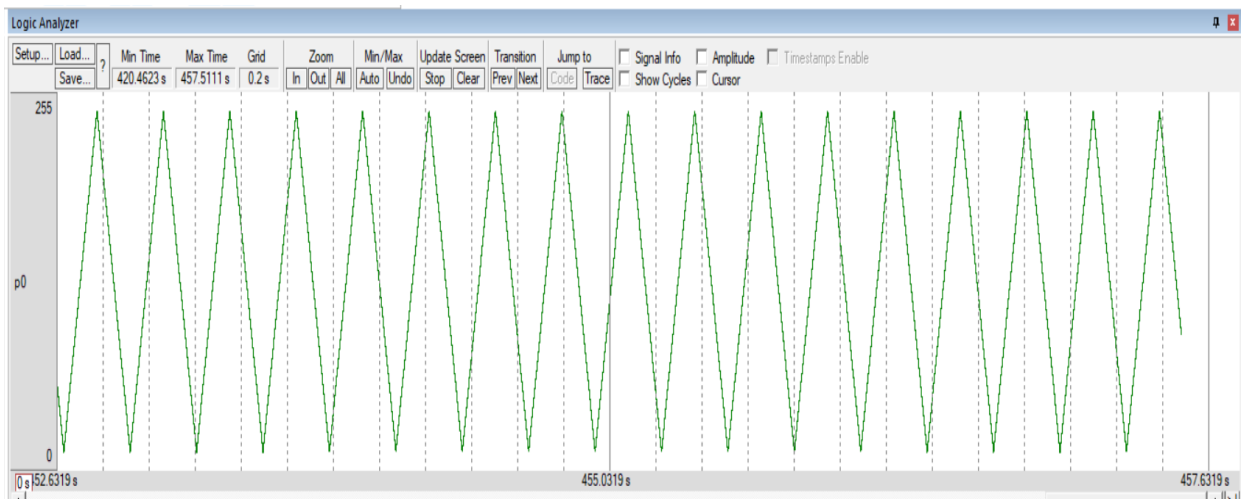
PROGRAM:**1.STAIRCASE:**

ADDRESS	OPCODES	LABELS	MNEMONICS	OPERANDS
0000	7400		MOV	A, #00H
0002	F590	BACK:	MOV	P1, A
0004	2410		ADD	A, #10H
0006	110C		ACALL	DELAY
0008	70F8		JNZ	BACK
000A	80FE	HERE:	SJMP	HERE
000C	7890	DELAY:	MOV	R0, #90H
000E	79FF	WAIT1:	MOV	R1, #0FFH
0010	7AFF	WAIT2:	MOV	R2, #0FFH
0012	DAFE	WAIT3:	DJNZ	R2, WAIT3
0014	D9FA		DJNZ	R1, WAIT2
0016	D8F6		DJNZ	R0, WAIT1
0018	22		RET	
			END	

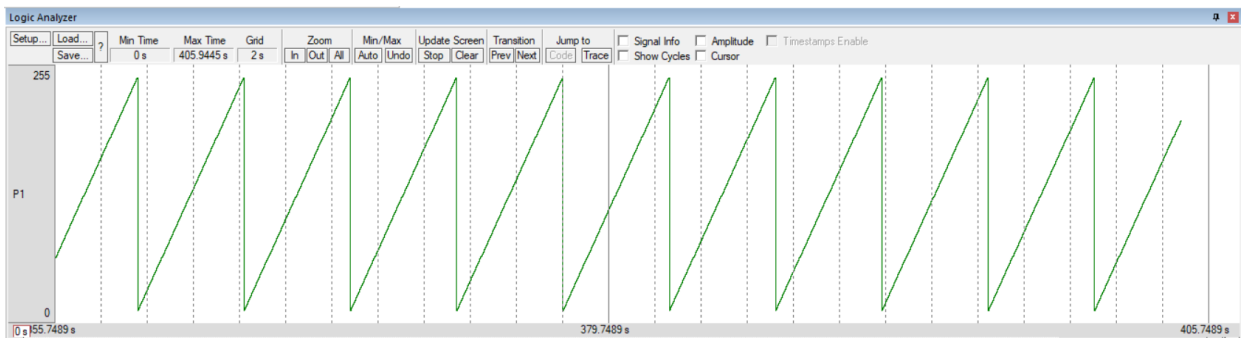
2.RAMP:

ADDRESS	OPCODE	LABEL	MNEMONICS	OPERAND
0000	7400		MOV	A,#00H
0002	F590	BACK:	MOV	P1,A
0004	04		INC	A
0005	110B		ACALL	DELAY
0007	70F9		JNZ	BACK
0009	80FE	HERE:	SJMP	HERE
000B	7840	DELAY:	MOV	R0,#40H
000D	7940	WAIT1:	MOV	R1,#40H
000F	7A80	WAIT2:	MOV	R2,#80H
0011	DAFE	WAIT3:	DJNZ	R2,WAIT3
0013	D9FC		DJNZ	R1,WAIT2
0015	D8F6		DJNZ	R0,WAIT1
0017	22		RET	

3.TRIANGULAR OUTPUT:



4.SAWTOOTH WAVE OUTPUT:



3.TRIANGULAR WAVE:

Address	Opcode	Label	Mnemonics	Operand
0000			ORG	00H
0000	7400	START:	MOV	A, #00H
0002	F580		MOV	P0, A
0004	7801		MOV	R0, #01H
0006		NEXT:		
0006	28		ADD	A, R0
0007	F580		MOV	P0, A
0009	1119		ACALL	DELAY
000B	B4FF04		CJNE	A, #0FFH, CHECK_ZERO
000E	78FF		MOV	R0, #0FFH
0010	80F4		SJMP	NEXT
0012		CHECK_ZERO:		
0012	B400F1		CJNE	A, #00H, NEXT
0015	7801		MOV	R0, #01H
0017	80ED		SJMP	NEXT
0019		DELAY:		
0019	7AFF		MOV	R2, #0FFH
001B		HERE:		
001B	DAFE		DJNZ	R2, HERE
001D	22		RET	
			END	

4.SAWTOOTH WAVE:

Address	Opcode	Label	Mnemonics	Operand
0000	7A50		MOV	A, #00H
0002	F590	BACK:	MOV	P1, A
0004	04		INC	A
0005	1109		ACALL	DELAY
0007	80F9		SJMP	BACK
0009	78FF	DELAY:	MOV	R0, #FFH
000B	791F	WAIT1:	MOV	R1, #1FH
000D	D9FE	WAIT2:	DJNZ	R1, WAIT2
000F	D8FA		DJNZ	R0, WAIT1
0011	22		RET	

RESULT:

The necessary interfacing circuit to interface the DAC with 8051 and the programs to generate different waveforms such as staircase, ramp, triangular, and sawtooth waves were successfully developed and executed.